

KAITAIA, New Zealand

WMO: 930120

Lat: 35.1328S

Lon: 173.2639E

Elev: 279

StdP: 14.55

Time Zone: 12.00 (NZL)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 41.4	(c) 43.4	(d) 36.7	(e) 32.3	(f) 46.1	(g) 38.5	(h) 34.7	(i) 47.3	(j) 34.5	(k) 56.4	(l) 32.1	(m) 55.9	(n) 5.3	(o) 100	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.2	76.8	66.6	75.2	66.1	73.7	65.4	70.1	72.5	68.8	71.7	67.6	71.0	12.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.2	108.5	71.2	67.9	103.5	70.0	66.5	98.5	69.1	34.3	72.3	33.2	71.7	32.2	70.9	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
							Mean		Standard Deviation		n=5 years		n=10 years		n=20 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.7	24.8	22.1	DB	37.9	80.7	1.6	2.1	36.8	82.2	35.9	83.4	35.0	84.6	33.9	86.1	
			WB	37.4	71.7	1.5	1.8	36.3	72.9	35.4	74.0	34.5	75.0	33.4	76.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	66.5	66.8	64.5	60.8	57.5	54.4	53.0	53.6	54.8	57.5	59.5	63.3
	DBStd	5.68	2.92	2.78	3.30	3.36	3.28	3.35	3.03	2.77	3.03	2.74	2.76	3.05
	HDD50	16	0	0	0	0	0	4	8	3	1	0	0	0
	HDD65	2260	17	13	47	129	235	317	371	353	307	234	166	71
	CDD50	3419	513	472	451	325	231	137	102	115	145	231	285	412
	CDD65	186	65	64	33	4	1	0	0	0	0	0	1	18
	CDH74	256	100	118	29	0	0	0	0	0	0	0	0	9
CDH80	2	0	2	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	9.8	9.7	8.9	8.8	8.6	9.8	8.7	10.2	10.6	11.2	11.5	10.6	9.3
Precipitation	PrecAvg	52.7	3.0	3.7	3.2	4.2	4.9	6.2	6.1	5.7	4.8	3.9	3.9	3.4
	PrecMax	73.0	11.3	14.7	8.9	13.0	12.8	13.9	14.1	12.2	9.2	8.4	8.7	7.3
	PrecMin	38.6	0.1	0.3	0.1	0.4	1.1	2.2	1.0	1.6	1.6	0.4	1.3	0.7
	PrecStd	9.5	2.3	3.2	2.1	2.4	2.4	2.5	2.6	2.3	1.7	1.9	1.9	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.5	79.2	76.8	71.6	67.8	64.6	62.6	62.7	65.4	67.9	70.9	75.7
		MCWB	67.5	66.9	67.7	62.8	63.4	62.6	58.8	57.7	60.6	60.3	62.5	65.1
	2%	DB	76.7	77.1	74.7	69.8	66.0	62.9	60.7	61.0	63.4	65.7	68.9	73.3
		MCWB	67.1	65.8	66.0	62.1	61.5	60.3	57.3	56.8	59.4	59.6	61.9	64.7
	5%	DB	75.2	75.2	72.9	68.2	64.7	61.6	59.4	59.9	61.8	64.3	67.4	71.7
		MCWB	66.7	65.6	65.3	61.9	60.4	58.6	56.0	56.1	57.2	59.0	61.2	63.9
	10%	DB	73.5	73.6	71.3	66.9	63.3	60.2	58.2	58.8	60.2	63.1	65.9	70.3
		MCWB	65.9	64.8	64.3	61.4	59.3	56.9	54.8	55.3	55.8	58.3	60.5	63.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.5	71.3	72.0	66.6	66.4	63.7	60.1	60.1	63.8	63.7	66.7	68.8
		MCDB	74.8	73.4	74.1	67.8	67.0	64.2	61.2	61.1	64.3	65.3	67.7	71.0
	2%	WB	69.9	69.1	70.2	65.6	63.8	61.7	58.5	58.8	60.2	62.2	64.5	67.6
		MCDB	72.5	72.8	72.1	67.3	64.8	62.5	59.8	60.1	62.0	63.9	66.5	70.1
	5%	WB	68.9	67.8	67.5	64.3	61.7	59.7	57.2	57.4	58.5	61.0	63.1	66.5
		MCDB	71.8	72.6	70.2	66.4	63.5	60.9	58.7	59.0	60.5	62.8	65.6	69.3
	10%	WB	67.9	66.5	65.6	62.7	60.0	57.9	55.7	56.1	57.0	59.7	61.7	65.3
		MCDB	71.2	71.2	69.6	65.6	62.4	59.5	57.4	58.1	59.4	61.9	64.6	68.4
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.2	12.7	11.3	10.4	9.9	9.7	9.6	10.0	10.5	11.0	12.6
		MCDBR	14.7	15.8	14.3	12.2	11.4	10.3	9.8	10.3	11.4	11.8	13.8	14.5
		MCWBR	6.9	6.9	7.0	6.8	7.6	7.4	6.8	7.0	7.0	7.6	8.1	7.4
	5% WB	MCDBR	9.6	12.4	9.6	9.2	8.4	7.7	7.5	8.1	8.2	9.2	9.4	10.1
MCWBR		5.7	6.8	6.4	6.7	6.7	7.1	6.4	6.6	6.1	7.0	7.1	6.3	
Clear-Sky Solar Irradiance	taub		0.344	0.340	0.328	0.317	0.311	0.308	0.303	0.312	0.335	0.329	0.334	0.340
	taud		2.573	2.620	2.645	2.665	2.653	2.641	2.658	2.621	2.528	2.560	2.567	2.599
	Ebn at Noon		314	309	302	287	272	263	272	285	294	309	316	317
	Edh at Noon		34	31	28	24	22	21	21	25	31	32	34	33
All-Sky Solar Radiation	RadAvg		2176	1922	1573	1153	818	673	729	999	1353	1740	1993	2072
	RadStd		126	132	99	66	41	39	44	56	68	108	128	166

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page